Customer Retention & Churn Analysis Dashboard

Future Interns · Data Science & Analytics · Task 2 | Prepared by: Sravya Rachuri

Executive Summary

This project analyzes customer churn in a telecommunications subscription dataset using Microsoft Power BI. An interactive business dashboard was developed to support data-driven retention decisions, bringing together customer counts, churn and retention KPIs, contract type, internet service, payment method, tenure, monthly charges, gender, and senior-citizen status.

The analysis covers 7,032 customers, of whom 1,869 churned and 5,163 did not churn. This corresponds to an overall churn rate of 26.58% and a retention rate of 73.42%.

Key findings reveal a strong link between contract type and churn: month-to-month customers churn at 42.71%, compared with 11.28% for one-year and 2.85% for two-year contracts. Senior citizens also show higher observed churn, and churned customers tend to have shorter tenure and higher average monthly charges. Findings describe associations only; no causal claims are made.

1. Business Objective

The primary objective of this project was to transform customer subscription data into meaningful business insights that could help stakeholders:

- Measure customer churn and retention using key performance indicators.
- Compare churn patterns across contract types and customer segments.
- Explore how internet service, payment method, tenure, and monthly charges relate to churn.
- Build an interactive dashboard that supports filtering and exploratory analysis.
- Recommend practical actions to improve customer retention.

2. Dataset Overview

Dataset Source

Telco Customer Churn Dataset - https://www.kaggle.com/datasets/blastchar/telco-customer-churn

Customer-level records covering demographics, subscribed services, contract and payment information, monthly and total charges, tenure, and a churn indicator. The Power BI model uses a cleaned table named Cleaned_Telco_Customer_Churn.

Dataset Summary

Metric	Value
Total Customers	7,032
Churned Customers	1,869
Retained Customers	5,163
Overall Churn Rate	26.58%
Primary Tool	Microsoft Power BI

Key fields used in the analysis:

Field	Meaning in this project
CustomerID	Identifier for an individual customer.
Churn	Whether the customer left the service (Yes) or did not churn (No).
Contract	Subscription contract type: month-to-month, one year, or two year.
InternetService	Internet service category: DSL, fiber optic, or no internet service.
PaymentMethod	Method used to pay for the service.
tenure	Length of time the customer has stayed with the company, in months.
MonthlyCharges	Recurring monthly amount charged to the customer.
Gender / Senior Citizen	Customer segment fields used for comparison.

3. Data Preparation & Analysis Method

The dashboard is built from the cleaned customer table in Power BI. Customer counts and DAX measures were used for churned customers, customers who did not churn, churn rate, and retention rate. Slicers for gender, contract, internet service, and senior-citizen status support segment-level exploration.

KPI Logic

KPI	Calculation concept
Total Customers	Count of customer records in the current filter context.
Active / Retained Customers	Count of records where Churn = No.
Churned Customers	Count of records where Churn = Yes.
Churn Rate	Churned Customers ÷ Total Customers.
Retention Rate	Active / Retained Customers ÷ Total Customers.

Note: selecting a bar or slicer cross-filters the other visuals, so KPI values must be interpreted in the active filter context. The unfiltered baseline is 7,032 total, 5,163 retained, and 1,869 churned customers.

4. Key Performance Indicators (KPIs)

The following KPIs were developed and displayed within the Power BI dashboard:

KPI	Value	Interpretation
Total Customers	7,032	Total customer records in the overall dashboard.
Active Customers	5,163	Customers with Churn = No.
Churned Customers	1,869	Customers with Churn = Yes.
Churn Rate	26.58%	Share of all customers who churned.
Retention Rate	73.42%	Share of all customers who did not churn.

Churn and retention percentages sum to 100%, and retained plus churned customers sum to the total customer count.

5. Business Analysis & Findings

5.1 Contract Type Analysis

Contract Type	Customers	Churned	Churn Rate
Month-to-month	3,875	1,655	42.71%
One year	1,472	166	11.28%
Two year	1,685	48	2.85%

Month-to-month customers have the highest churn rate, and the observed rate falls substantially for one-year and two-year contracts.

Business Interpretation

Customers without a long-term commitment may be more likely to leave. The dashboard shows an association, not proof that contract length itself causes retention.

5.2 Gender-Based Churn Analysis

Gender	Total Customers	Churned	Churn Rate
Female	3,483	939	26.96%
Male	3,549	930	26.20%

Female and male customers have similar churn rates, differing by about 0.76 percentage points.

Business Interpretation

Gender does not appear to be a strong differentiator of churn, so retention efforts need not be split by gender.

5.3 Senior Citizen Analysis

Segment	Total Customers	Churned	Churn Rate
Senior citizen: Yes	1,142	476	41.68%
Senior citizen: No	5,890	1,393	23.65%

The senior-citizen segment has a notably higher observed churn rate than the non-senior segment.

Business Interpretation

This segment merits further investigation into contract type, service experience, and pricing. The comparison does not establish a cause.

5.4 Customer Tenure Analysis

Churn Status	Average Tenure (Overall View)
Churn = Yes	Approximately 18 months

Churn Status	Average Tenure (Overall View)
Churn = No	Approximately 38 months

In the overall dashboard view, churned customers have an average tenure of about 18 months, compared with about 38 months for customers who stayed.

Business Interpretation

The early customer lifecycle is worth investigating. Onboarding and early support may help reduce churn.

5.5 Average Monthly Charges

Churn Status	Average Monthly Charges (approx.)
Churn = Yes	74
Churn = No	61

Customers who churned show higher average monthly charges than customers who stayed.

Business Interpretation

This may indicate a relationship between bill level and churn, but the chart does not prove that higher charges caused customers to leave.

5.6 Internet Service Analysis

The dashboard compares churn counts for fiber optic, DSL, and customers with no internet service. In the overall view, fiber optic has the largest visible churn count.

Business Interpretation

The chart uses customer counts, so it does not establish that fiber optic has the highest churn rate. Churned customers divided by all customers within each category should be calculated before ranking risk.

5.7 Payment Method Analysis

The payment-method chart compares churned and retained customer counts across electronic check, mailed check, bank transfer, and credit card. Electronic check has the largest visible churn count in the overall view.

Business Interpretation

A rate-based comparison is needed before concluding that electronic-check users carry the highest churn risk.

6. Power BI Dashboard

An interactive Power BI dashboard titled Customer Retention & Churn Analysis Dashboard was developed to allow stakeholders to explore churn dynamically.

Dashboard Components

- Total Customers KPI
- Active Customers KPI

- Churned Customers KPI
- Churn Rate KPI
- Retention Rate KPI
- Customer Churn Distribution (donut chart)
- Contract Type vs Customer Churn (stacked bar chart)
- Internet Service vs Customer Churn (stacked bar chart)
- Payment Method vs Customer Churn (stacked bar chart)
- Customer Tenure Distribution chart
- Average Monthly Charges by Churn chart
- Slicers: Gender, Contract, Internet Service, Senior Citizen Status

The dashboard enables users to analyze churn across multiple dimensions without requiring technical expertise. Values shown after selecting a slicer or chart bar represent the filtered segment, not the full dataset.

7. Strategic Recommendations

Based on the analysis, the following recommendations are proposed:

1. Prioritize Month-to-Month Customers
Test suitable loyalty benefits, renewal incentives, or longer-term plan options for customers who are open to them.

2. Strengthen Early Retention
Improve onboarding and provide timely support during the early customer lifecycle, then monitor churn by tenure band.

3. Investigate Senior-Citizen Churn
Compare contracts, internet service, monthly charges, and support experience within this segment before choosing targeted interventions.

4. Review Perceived Value and Pricing
Assess whether customers with higher monthly charges receive services and support that meet their expectations.

5. Evaluate Service and Payment Segments with Rates
Calculate category-level churn rates for internet service and payment method rather than relying on churn counts alone.

6. Monitor KPIs Regularly
Track churn rate, retention rate, and segment-level results over time to evaluate whether interventions are working.

8. Power BI Dashboard

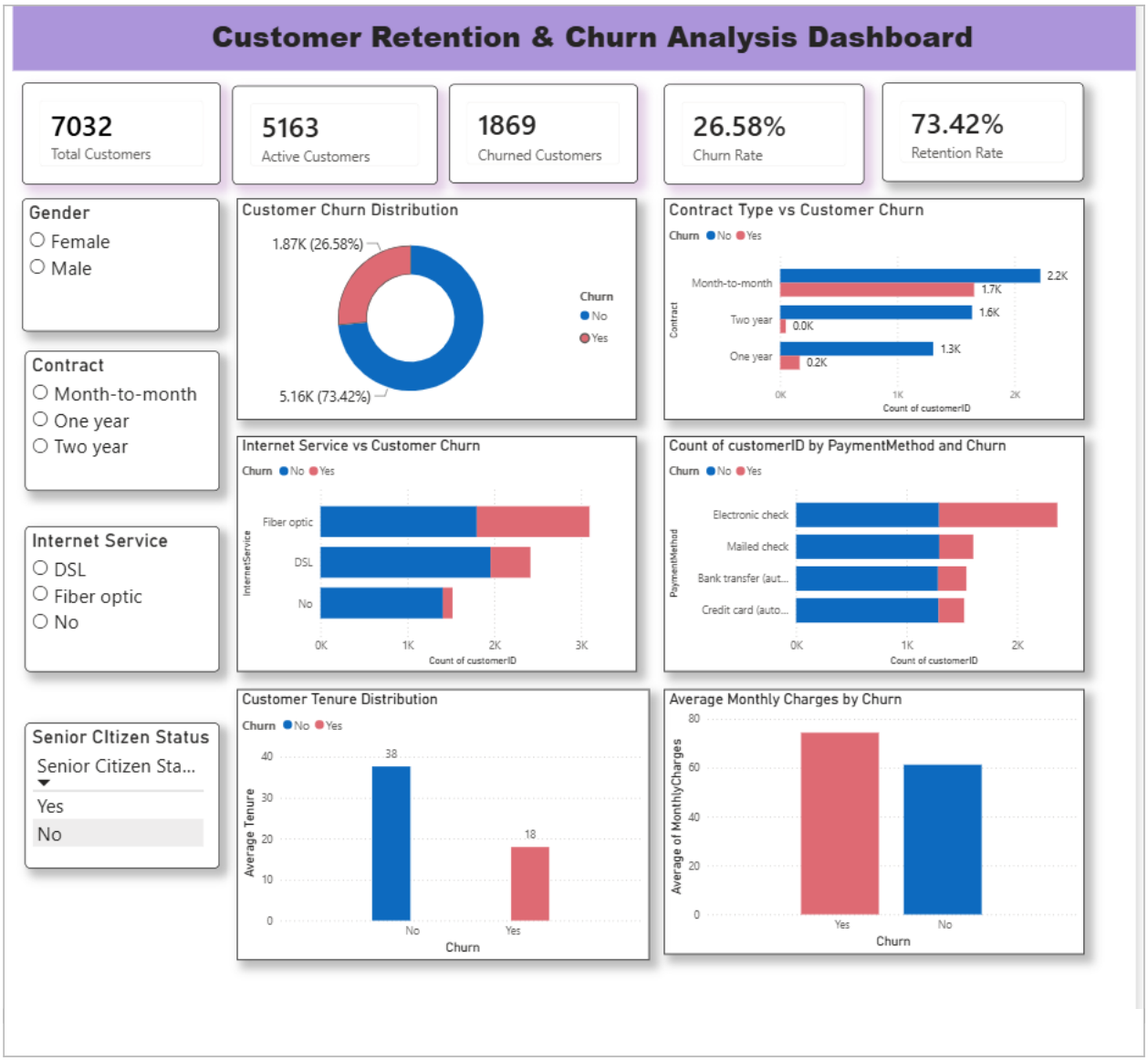


Figure: Interactive Power BI Dashboard showing churn, retention, contract, tenure, and charge metrics.

9. Conclusion

This project demonstrates how customer subscription data can be transformed into actionable business intelligence to support retention planning.

Through data preparation, KPI development, exploratory analysis, and dashboard development, valuable insights were uncovered. The most prominent result is the higher churn rate among month-to-month customers (42.71%) compared with one-year (11.28%) and two-year customers (2.85%). The analysis also identifies higher observed churn among senior citizens, similar churn rates by gender, shorter average tenure among churned customers, and higher average monthly charges among churned customers.

Further analysis should calculate churn rates for internet-service and payment-method segments and investigate the factors associated with churn before drawing causal conclusions. The resulting Power BI dashboard provides decision-makers with a practical tool for monitoring customer behavior and prioritizing evidence-based retention initiatives.